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Feasibility Study of a Novel Stent-Based Low-Level Laser Therapy Device for Reducing Severe Radiation-Induced Oral Mucositis

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ABSTRACT

Objectives: Radiation-induced oral mucositis (RIOM) is a severe complication of radiotherapy (RT) for head and neck cancers, impairing quality of life and treatment adherence. Low-level laser therapy (LLLT) shows promise for reducing RIOM severity, but operator-dependent devices limit practicality. This feasibility study assessed the safety and efficacy of a novel intraoral stent-based LLLT device designed for consistent laser delivery.

Materials and Methods: This prospective, single-arm, multicenter study enrolled 15 patients receiving curative RT for head and neck cancers. The stent-based device delivered laser light to 43 oral cavity points at three wavelengths (670, 830, and 910 nm) during 20-min sessions, five times weekly. The primary end point was the incidence of Grade 3 or higher RIOM (WHO oral mucositis scale). Pain severity was a secondary end point (Numeric Pain Rating Scale [NPRS]).

Results: Grade 3 or higher RIOM occurred in 6.6% of patients, significantly lower than historical rates (30%–40%). No patients reported severe pain (NPRS $\geq$ 8). The device was well-tolerated with no adverse effects, and all patients completed RT as planned.

Conclusions: The intraoral stent-based LLLT device is a safe and potentially effective intervention for reducing severe RIOM. Randomized trials are warranted to confirm these findings.

Trial Registration: This study was a prospective, single-arm feasibility trial and was not registered in a clinical trial registry, as it did not meet the criteria for mandatory registration at the time of its initiation.

1 | Introduction

Head and neck cancers comprise a group of malignancies originating in the facial and neck regions, with an annual incidence of approximately 5000 cases in Korea, accounting for about 2% of all cancers [1]. Nearly half of these patients present with advanced-stage disease at diagnosis, necessitating multimodal treatment approaches that include surgery, radiotherapy (RT), and chemotherapy [2, 3]. Among these, RT is indispensable, serving both as a definitive treatment for organ

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preservation and as an adjuvant modality to reduce recurrence after surgery.

Radiation-induced oral mucositis (RIOM) is a frequent and clinically significant side effect of RT for head and neck cancers [4]. Severe mucositis (Grade 3 or higher) occurs in approximately 29%–63% of cases and significantly impairs patients' quality of life by affecting essential functions such as chewing, swallowing, and speaking [4–7]. Furthermore, severe mucositis increases the risk of secondary infections, prolongs hospital stays, elevates healthcare costs, and most critically, can necessitate treatment interruptions that can compromise oncologic outcomes.

Despite various preventive and therapeutic strategies, effective solutions for RIOM remain limited. Low-level laser therapy (LLLT) has emerged as a promising intervention for managing RIOM. LLLT may work by improving mitochondrial function, leading to increased energy production (ATP) and reduced oxidative stress. It also supports tissue repair by stimulating fibroblasts, enhancing collagen production, and reducing inflammation [8]. Based on robust evidence from randomized controlled trials and meta-analyses [9–18], the Multinational Association of Supportive Care in Cancer/International Society of Oral Oncology (MASCC/ISOO) guidelines recommend LLLT as a Level 1 or 2 intervention for RIOM prevention [19]. Long-term studies have also mitigated concerns regarding LLLT-related tumor recurrence, underscoring its safety [20–22].

Traditional handheld LLLT devices, while effective, have practical limitations. Each treatment point requires laser application for several seconds to minutes, and multiple points often need treatment in a single session [9–18]. This method can be time-consuming, challenging for healthcare providers to administer consistently, and uncomfortable for patients due to prolonged sessions. To overcome these limitations, the intraoral stent-based LLLT device was developed. By delivering laser light to 43 points simultaneously, this device ensures uniform irradiation of a broader area, reduces variability in treatment application, and minimizes patient discomfort. Furthermore, its straightforward design, which requires patients to hold the stent in place for 20 min per session, integrates seamlessly into clinical workflows.

This exploratory clinical trial aims to assess the safety and feasibility of this newly developed LLLT approach for preventing severe RIOM in patients undergoing RT for head and neck cancers. The study assesses whether this innovative intervention can reduce the incidence of Grade 3 or higher RIOM while maintaining patient safety and treatment compliance.

2 | Materials and Methods

2.1 | Study Design

This prospective, open-label, single-arm, multicenter feasibility study evaluated the safety and efficacy of an intraoral stent-based LLLT device for preventing RIOM. The study spanned from September 2023 to May 2024, with approvals from the Institutional Review Boards of the three participating institutions—Samsung Medical Center, Seoul National University Bundang Hospital, and the National Cancer Center (IRB number: 2023-05-129). It adhered to the principles of the Declaration of Helsinki and the Good Clinical Practice guidelines, ensuring compliance with all ethical and regulatory standards.

Fifteen patients were proportionally enrolled: six each at Samsung Medical Center and Seoul National University Bundang Hospital, and three at the National Cancer Center. The objective was to determine if the intraoral stent-based LLLT could reduce the incidence of severe RIOM compared to historical data, while also monitoring for any adverse effects.

2.2 | Patient Selection

Eligible participants were adults aged between 19 and 70 years, diagnosed with cancers affecting the oral cavity, oropharynx, nasopharynx, or salivary glands. All participants were scheduled to receive RT with a cumulative dose of at least 50 Gy to the oral cavity. Written informed consent was obtained from all participants before inclusion.

Exclusion criteria included patients scheduled for palliative RT, those unable to wear the intraoral stent due to restricted mouth opening, pregnant individuals, and those with uncontrolled connective tissue disorders. Additional reasons for exclusion were a history of phototherapy within the past 6 weeks, use of photosensitizing medications, or conditions likely to interfere with study participation, such as severe psychological disorders.

2.3 | Stent-Based LLLT

The intraoral stent-based LLLT device (PMD-PHM2000, PTECH Co. Ltd., Republic of Korea) (Figure 1) is engineered to simultaneously deliver laser light to 43 designated points in the oral cavity. This includes 12 points on the upper surface, 19 on the lower surface, and 12 on the sides, covering the recommended areas (Figure 2) [18]. This innovative design ensures uniform laser exposure across the targeted areas, thereby enhancing therapeutic efficacy. The device operates at three wavelengths—670 nm (AlGaInP, 1.34 mW), 830 nm (GaAs, 2.14 mW), and 910 nm (InGaAs, 0.93 mW)—in pulse mode, delivering an energy density of 6.7 J/cm^2 and a total energy output of 227 J during each 20-min session.

Unlike traditional LLLT devices that require handheld probes to treat each point individually for 30–60 s, this stent-based system simplifies the procedure by delivering laser light to multiple points simultaneously. This design significantly reduces treatment time and eliminates the need for continuous operator intervention, thereby improving procedural consistency and minimizing variability in treatment delivery.

Laser therapy was started during the second week of RT and administered five times per week until the completion of RT. Each treatment session lasted 20 min. Before each session, patients were instructed to rinse their mouths with lukewarm



FIGURE 1 | Intraoral stent with laser probes. (A) Laser probe. (B) Laser applicator.

water. A sterile cover was then placed over the stent, which was positioned in the oral cavity. The stent was removed after treatment, and the patients were instructed to rinse their mouths again with lukewarm water.

2.4 | Evaluation

The primary end point was the incidence of Grade 3 or higher RIOM, defined as oral ulcers restricting dietary intake to liquids, assessed using the World Health Organization (WHO) oral mucositis scale. A previous randomized study demonstrated that LLLT significantly reduces the incidence of Grade 3 or higher mucositis to 6.5% compared to 40.5% in the control group [20]. This study aimed to achieve an incidence of Grade 3 or higher mucositis in no more than two participants (≤ 13.3%).

The secondary end point was pain severity, assessed using the Numeric Pain Rating Scale (NPRS). In the prior randomized study, the incidence of pain scores of 5–7 and 8–10 in the laser-treated groups was 19.1% and 2.1%, respectively, compared to 21.9% and 14.9% in the control group [20]. In this study, it was anticipated that fewer than 5% of participants (≤ 1 patient) would report severe pain (NPRS 8–10) during treatment.

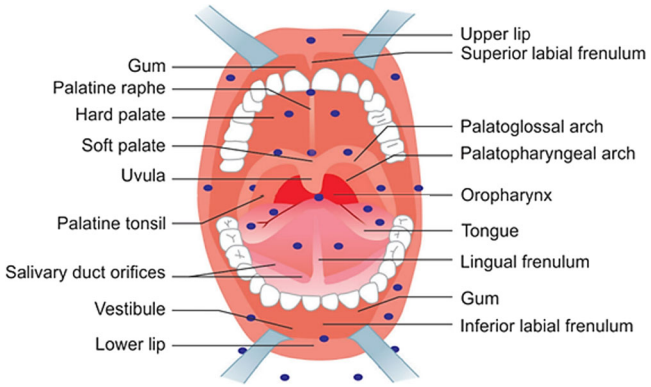


FIGURE 2 | Recommended laser application points for low-level laser therapy. From Kuhn-Dall'magro et al. [18].

Weekly assessments were conducted during RT to evaluate the severity of oral mucositis and pain levels. These assessments included one-on-one interviews with study investigators, during which the severity of RIOM was evaluated. Based on the degree of mucositis observed, symptomatic management, such as the use of analgesics and anti-inflammatory medications, was provided in accordance with institutional protocols or guidelines.

Adverse events and device-related complications were recorded throughout the study. A final evaluation was conducted 4 weeks after the completion of RT to consolidate the outcomes and confirm the safety and feasibility of the intervention.

All evaluations were performed directly by radiation oncology faculty specializing in head and neck cancer at tertiary referral hospitals. Photographic documentation was not included in the study protocol, as standardization across multiple institutions (e.g., lighting conditions, camera resolution, and patient positioning) was considered difficult and potentially a source of bias. Instead, the investigators discussed the WHO oral mucositis scale criteria in advance and reached consensus on their application, ensuring standardized and reliable assessments across institutions.

3 | Results

Baseline characteristics are described in Table 1. A total of 15 patients undergoing curative RT for head and neck cancers were enrolled from three institutions. The cohort included patients with cancers of the oral cavity ($n = 5$), oropharynx ($n = 6$), nasopharynx ($n = 1$), salivary glands ($n = 2$), and one patient with a malignancy of unknown origin (MUO) involving the right neck. All patients received a minimum radiation dose of 50 Gy to the oral cavity. The average total radiation dose delivered was 6074 cGy, with a mean maximum dose to the oral cavity of 6241 cGy. All patients completed the planned treatment protocol without significant deviations.

TABLE 1 | Baseline characteristics of patients.

Characteristics		Number of patients
Age, years	Median (range)	55 (45–66)
Gender	Male	12 (80.0%)
	Female	3 (20.0%)
Tumor site	Oral cavity	5 (33.3%)
	Oropharynx	6 (40.0%)
	Nasopharynx	1 (6.7%)
	Salivary gland	2 (13.3%)
	MUO	1 (6.7%)
RT aim	Definitive	9 (60.0%)
	Postoperative	6 (40.0%)
Concurrent chemotherapy	Yes	7 (46.7%)
	No	8 (53.3%)
Total dose, cGy	Mean (range)	6074.7 (5700–6750)
Oral cavity$_{\text{max dose}}$, cGy	Mean (range)	6242.1 (5051–7097)
Oral cavity$_{\text{mean dose}}$, cGy	Mean (range)	2928.8 (1058.3–4857.8)

Abbreviations: MUO, malignancy of unknown origin; RT, radiotherapy.

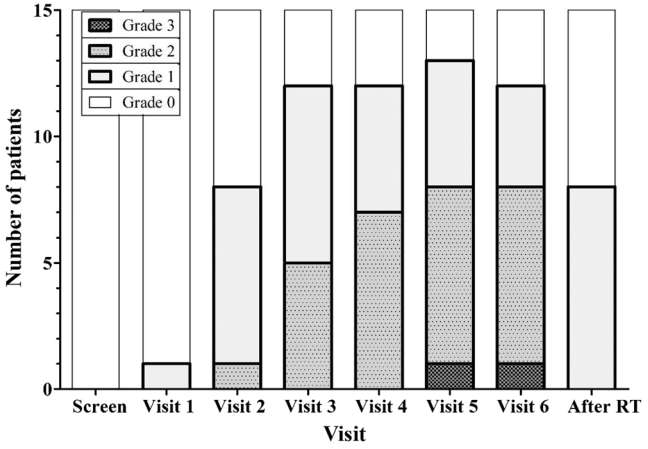


FIGURE 3 | Changes in World Health Organization (WHO) oral mucositis grades during radiotherapy. Values represent the maximum mucositis grade observed at each week of treatment.

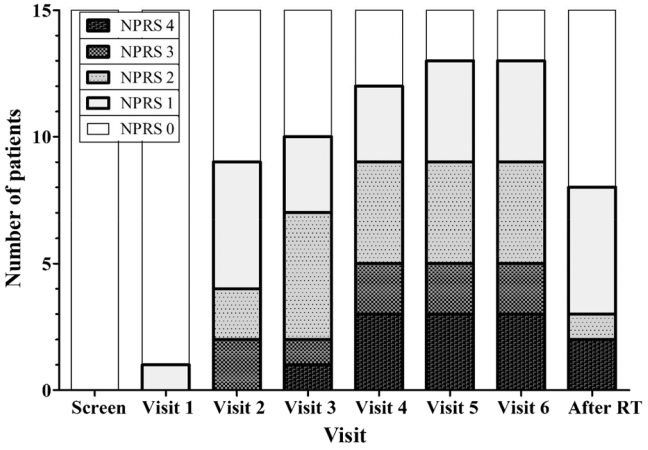


FIGURE 4 | Changes in Numeric Pain Rating Scale (NPRS) scores during radiotherapy. Values represent the maximum pain score reported by patients each week.

The median grade of RIOM peaked during Weeks 5–6 of RT (Figure 3). The primary end point, the incidence of Grade 3 or higher RIOM, was observed in only one patient (6.6%), who exhibited Grade 3 RIOM starting at Week 5 of RT.

Pain levels, assessed using the NPRS, showed that no patients reported moderate (NPRS = 5–7) or severe pain (NPRS ≥ 8) at any point during the study (Figure 4). Four patients reported pain levels up to NPRS 4, while the remaining patients reported pain levels of 3 or below throughout the study period.

Although five patients experienced treatment interruptions of 1–5 days, all completed their planned RT schedules successfully. The intraoral stent-based LLLT device was well-tolerated and did not interfere with the standard treatment regimen. No

additional adverse effects attributable to the LLLT device were reported, underscoring its safety and feasibility.

4 | Discussion

This study evaluated the safety and feasibility of a novel intraoral stent-based LLLT device for preventing severe RIOM in patients undergoing RT for head and neck cancers. The findings indicated a substantial reduction in the incidence of Grade 3 or higher RIOM, with only 6.6% of patients affected, compared to the historical rates of 30%–40%. Additionally, pain severity remained low throughout the study, with no patients reporting severe pain (NPRS $\geq$ 8), and the majority experiencing pain levels of NPRS 3 or below. Importantly, the device was well-tolerated, with no reported adverse effects or significant disruptions to RT schedules, highlighting its potential for seamless integration into clinical practice. These results affirm the feasibility of this intervention and suggest it may overcome some of the challenges associated with conventional LLLT methods.

The observed reduction in RIOM incidence aligns with prior research suggesting that LLLT may mitigate mucositis severity through mechanisms such as increased ATP production and reduced oxidative stress [8, 20]. However, LLLT parameters, including wavelength, energy density, and total energy delivered, vary considerably across studies. Most previous research has utilized wavelengths ranging from 632 to 970 nm, with energy densities of 2–12 J/cm^2, and total energy between 14.88 and 72 J [15–17]. Despite the variability in protocols, there are currently no universally accepted guidelines for optimal LLLT parameters [15]. In this study, we utilized wavelengths of 670, 830, and 910 nm in pulse output, delivering an energy density of 6.7 J/cm^2 to 43 points over a 20-min session, resulting in a total energy delivery of 227 J. Although the total energy delivered was relatively high, the intraoral stent allowed for uniform and stable irradiation over a wide area, ensuring safety and eliminating issues such as burns or mucosal irritation. This highlights the advantage of the stent-based device in providing consistent and safe LLLT delivery across the targeted areas.

The low pain levels observed in this study further highlight the potential benefits of LLLT in improving patient comfort during RT. Consistent with prior research, LLLT's ability to modulate nociceptive signaling and reduce local inflammation likely contributed to the absence of moderate or severe pain in the study population [8, 20]. This is particularly significant, as effective pain management is crucial for ensuring adherence to RT schedules, which is essential for achieving optimal oncologic outcomes.

All patients in the study completed their planned RT regimens, despite minor treatment interruptions in five patients. This high compliance rate reflects the practicality of the stent-based device, which simplifies the LLLT procedure and integrates seamlessly into standard RT workflows. No device-related adverse effects were observed, supporting the safety of the stent-based LLLT device and its potential for broader clinical adoption. Recent studies have also shown no increase in cancer recurrence among patients treated with LLLT for mucositis prevention [20–22]. However, conflicting evidence from animal models about the effects of PBM on tumor behavior underscores the need for caution. Clinicians should inform patients about both the potential benefits and risks of this approach [23, 24].

Despite these promising findings, this study has important limitations. First, the small sample size, inherent to the feasibility trial design, substantially limits the generalizability of the results. Second, the absence of a control group prevents definitive conclusions regarding the comparative efficacy of the device. Third, this study did not assess long-term outcomes, including impacts on patient-reported quality of life and oncologic end points. Future studies with larger, randomized cohorts are necessary to validate these findings and further explore the long-term benefits of stent-based LLLT.

In conclusion, this feasibility trial demonstrates that the intraoral stent-based LLLT device is a safe, effective, and practical intervention for reducing the incidence of severe RIOM and improving patient comfort during RT for head and neck cancers. By simplifying the application of LLLT and ensuring consistent delivery, this device addresses key limitations of traditional approaches, paving the way for its integration into standard clinical practice. Further research is warranted to confirm these findings and establish the device's role in optimizing supportive care for RT patients.

5 | Conclusion

This feasibility study indicates that the intraoral stent-based LLLT device could be a safe and potentially effective intervention for reducing the incidence of severe RIOM in patients undergoing RT for head and neck cancers. Patients reported a low incidence of Grade 3 or higher RIOM, experienced minimal pain, and reported no device-related adverse effects. These findings underscore the potential of this device to enhance supportive care in RT settings. However, larger, randomized trials are necessary to confirm these results and to explore the long-term benefits of the device.

Ethics Statement

This study was approved by the Institutional Review Boards of all three participating institutions—Samsung Medical Center, Seoul National University Bundang Hospital, and the National Cancer Center (IRB No. 2023-05-129). The study was conducted in accordance with the principles of the Declaration of Helsinki and Good Clinical Practice guidelines, ensuring full compliance with applicable ethical and regulatory standards.

Consent

Written informed consent was obtained from all participants before enrollment. The consent process included a detailed explanation of the study purpose, procedures, potential risks, and benefits.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are stored in an institutional repository and will be shared upon request to the corresponding author.

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